SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

HA CRM - Multi-Tenant Customer Relationship Management System Optimized for (PHP / MySQL / JS)

Project Name:	HA CRM Panel
Document Version:	1.0.0
Date:	September 2026
Target Platform:	Shared Web Hosting (cPanel / LAMP / PHP 8.x / MySQL)
Core Architecture:	Non-VPS Modular Monolith (MVC / Native PHP Router)
Status:	Approved Technical Specifications

1. Introduction & System Scope

1.1 Purpose

This Software Requirements Specification (SRS) provides an exhaustive functional, UI/UX, database, and technical layout for 'HA CRM'—a comprehensive Customer Relationship Management system designed specifically for high-throughput operational efficiency on Shared Web Hosting (Non-VPS) infrastructure without AI overhead or dependencies.

1.2 Scope

HA CRM streamlines end-to-end sales pipelines, customer databases, lead automation, task assignments, quote/invoice billing, customer support ticketing, and detailed analytics. The system operates entirely on native web standards (PHP, MySQL, HTML5, JavaScript, AJAX), delivering zero-latency operations, robust role-based access control (RBAC), and strict transactional security without requiring VPS servers, Node.js daemons, or third-party AI APIs.

1.3 Technical Constraints & Non-VPS Environment

- **Hosting Infrastructure:** Standard Apache/LiteSpeed Shared Hosting (cPanel / DirectAdmin / Plesk).
- **Server Language & Runtime:** PHP 8.1 / 8.2 with PDO extension and MySQLi enabled.
- **Database Engine:** MySQL 8.0 or MariaDB 10.5+ with InnoDB storage engine.
- **Client-Side Tech Stack:** HTML5, CSS3 (Tailwind CSS via CDN / Bootstrap 5), JavaScript (Vanilla ES6 / Axios / DataTables.js).
- **Server Restrictions Complied With:** No root/SSH background daemons, no Python runtime requirements, no WebSockets (uses HTTP Long-Polling/AJAX for notifications), memory footprint < 64MB per PHP execution.

2. Global Layout & Screen UI Architecture

2.1 Universal Dashboard Layout

All screens in HA CRM follow a consistent responsive 3-column layout structure engineered for desktop and tablet productivity:

- **Left Navigation Sidebar:** Collapsible vertical panel containing brand identifier, user quick profile badge, and grouped navigation menus with active state highlights.
- **Top Utility Bar (Header):** Contains Global Search Bar (Omnibox), Quick Action Button ('+ Add New'), Notification Bell Center, Active Task Counter, and User Profile Dropdown.
- **Main Canvas Workspace:** Central dynamic area containing Page Header (Breadcrumbs, Page Title, Action Buttons) and Content Grid Cards / Data Tables.
- **Right Contextual Slide-Over Drawer:** Off-canvas panel that opens on clicking any lead, deal, or contact without navigating away from the current workspace.

2.2 Comprehensive Sidebar Navigation Hierarchy

The left sidebar is categorized logically to grant quick access to core CRM domains based on role permissions:

Sidebar Menu Group	Sub-Menu Module	Screen Purpose & Content
Dashboard	Executive & Sales Overview	High-level KPI metrics, pipeline progress, conversion graphs, upcoming tasks.
Lead Management	All Leads / Unassigned Queue	Lead table with status filters, lead capture form, lead assignment rules.
Sales Pipeline	Deals Kanban / List View	Visual drag-and-drop Kanban board across deal stages, total pipeline value.
Contact Directory	Contacts & Accounts	Individual customer records, company organization profiles, interaction history.
Activities & Tasks	Task List & Calendar	Call logs, meeting scheduler, follow-up reminders, team activity feed.
Invoicing & Billing	Quotes & Invoices	Quotations builder, tax invoice generator, payment status tracking, PDF download.
Customer Support	Tickets & Helpdesk	Support ticket queue, SLA tracking, internal agent notes, customer response editor.
Reports & Analytics	Sales & Team Reports	Revenue forecast reports, agent performance benchmarks, lead source ROI.
System Settings	Company & User Settings	Roles & permissions, Lead stages setup, Custom fields,

	Email SMTP configuration.

3. Module-by-Module Screen Content & Field Requirements

3.1 Lead Management Module

A. Screen Layout & Content

The Lead Management screen features a top KPI summary strip (Total Leads, New Today, Qualified Rate, Unassigned Count), followed by an advanced filter toolbar (Search by Name/Email, Lead Source Filter, Lead Score, Assigned Rep Filter). Below the toolbar is a rich Data Table with server-side pagination, inline field editing, and quick action buttons.

B. Required Form Fields (Lead Creation / Drawer View)

- **Basic Information:** Salutation, First Name*, Last Name*, Job Title, Organization/Company Name.
- **Contact Details:** Primary Email*, Secondary Email, Phone Number*, Mobile Number, Website URL.
- **Lead Categorization:** Lead Status (New, Contacted, Qualified, Unqualified, Lost), Lead Source (Website Form, Referral, Cold Call, Paid Ad, Event), Lead Industry.
- **Qualification & Value:** Estimated Lead Value (₹), Lead Score (1-100), Priority (Low, Medium, High, Urgent).
- **Assignment & System:** Assigned Sales Representative (Dropdown), Assigned Team/Department, Next Follow-Up Date & Time, Internal Notes.

3.2 Sales Pipeline & Deals Kanban Module

A. Screen Layout & Content

Features an interactive Drag-and-Drop Kanban Board with customizable pipeline columns (e.g., Lead In, Meeting Scheduled, Proposal Sent, Negotiation, Closed Won, Closed Lost). Each column header displays total deal count and aggregated monetary value. Kanban cards display Deal Title, Company, Assigned Rep Avatar, Expected Close Date, Deal Amount, and a visual flag for overdue tasks.

B. Required Form Fields (Deal Details & Drawer)

- **Deal Identification:** Deal Title*, Pipeline Name, Pipeline Stage*, Associated Contact (Lookup), Associated Account/Company (Lookup).
- **Financial Metrics:** Deal Amount (₹)*, Discount Amount (₹), Tax Rate (%), Net Deal Value (₹), Expected Close Date*.
- **Pipeline Tracking:** Win Probability (0-100%), Loss Reason (Dropdown: Price, Competitor, Feature Gap, No Response), Deal Owner.

3.3 Contact & Account Directory Module

A. Screen Layout & Content

A dual-tabbed module dividing individual contacts from company accounts. Includes a split-screen view: Left list panel (30% width) for browsing contacts/accounts; Right detail panel (70% width) showing full profile details, communication timeline, linked deals, open invoices, and activity logs.

B. Required Form Fields

- **Contact Record:** Full Name*, Direct Email*, Direct Phone*, LinkedIn Profile URL, Billing Address, Shipping Address, Preferred Contact Method (Email/Phone/WhatsApp).
- **Account Record:** Company Name*, Registration Number, Industry Sector, Annual Revenue, Employee Count, Parent Company, Main Office Phone, Office Address, Tax ID.

3.4 Activities, Tasks & Calendar Module

A. Screen Layout & Content

Includes a full interactive Monthly/Weekly/Daily Calendar view along with a List View grouped by 'Overdue', 'Due Today', 'Upcoming', and 'Completed'. Users can log calls, schedule client meetings, create follow-up reminders, and assign tasks to teammates.

B. Required Form Fields

- **Activity Data:** Activity Type (Call, Meeting, Email, Task, Demo)*, Subject/Title*, Related Entity (Lead / Deal / Contact), Due Date & Time*, Duration (Mins), Status (Pending, In Progress, Completed, Cancelled), Priority, Detailed Description, Assigned To.

3.5 Invoicing & Quotation Module

A. Screen Layout & Content

Includes a line-item quotation and invoice generator with live tax/discount recalculations, PDF generation engine (FPDF/TCPDF via native PHP), and payment status badges (Draft, Sent, Paid, Overdue, Partially Paid).

B. Required Form Fields

- **Document Header:** Invoice/Quote Number (Auto-generated), Reference Deal, Issue Date*, Due Date*, Currency (USD, EUR, GBP, INR).
- **Line Items Grid:** Item Name*, Description, Quantity*, Unit Price*, Tax Rate (%), Item Total (₹).
- **Totals & Notes:** Subtotal, Global Discount (₹/%), Tax Total, Grand Total*, Terms & Conditions, Payment Bank Details, Customer Notes.

4. Internal Connections, Triggers & Automated Workflows

4.1 Entity Relationship Diagram & Data Connections

HA CRM connects all modules seamlessly using foreign key relationships in MySQL to ensure dynamic data flow across the system:

- **Lead to Deal Conversion:** Converting a qualified Lead automatically creates a Contact Record, creates an Account Record (if company is listed), and opens a new Deal in Stage 1 of the Kanban board.
- **Deal to Invoice Connection:** Products/Services quoted inside a Deal can be converted with 1-click into a formal Tax Invoice populated with customer billing details.
- **Contact to Support Ticket Connection:** When a Contact submits a helpdesk request, the ticket is linked to their CRM timeline, allowing agents to see past purchases and deal sizes during support resolution.

4.2 Cron-Based Workflow Automations

Automation runs on standard cPanel scheduled Cron jobs (`cron.php` running every 5 or 15 minutes) to perform background operational tasks without needing Python or Node background processes:

Trigger Event	System Connection / Action Taken	Execution Mechanism (Non-VPS)
New Web Form Lead Submission	Validates payload, applies Round-Robin assignment to online sales reps, triggers SMTP email notification to rep.	PHP POST Endpoint (`api/lead_capture.php`)
Deal Stage Moved to 'Closed Won'	Generates draft invoice, updates rep monthly quota meter, sends welcome email to client.	PHP Controller Event Hook (`DealController.php`)
Invoice Unpaid After Due Date	Marks invoice status as 'Overdue', creates follow-up task for account manager, sends polite payment reminder email.	cPanel Scheduled PHP Cron (`cron/check_overdue.php`)
Task Due in 30 Minutes	Pushes browser desktop alert and internal bell notification to assigned user.	AJAX Long-Polling Header Script (`notification_poll.php`)

5. Optimized Database Schema (MySQL / InnoDB)

5.1 Core Database Tables Schema

Below is the production-ready SQL database schema designed for maximum speed and zero foreign-key deadlocks on shared web hosting:

```
-- HA CRM Production Database Schema for Non-VPS MySQL

CREATE TABLE `users` (
  `id` INT AUTO_INCREMENT PRIMARY KEY,
  `full_name` VARCHAR(100) NOT NULL,
  `email` VARCHAR(100) NOT NULL UNIQUE,
  `password_hash` VARCHAR(255) NOT NULL,
  `role` ENUM('Admin', 'Sales Manager', 'Sales Rep', 'Support Agent') NOT NULL DEFAULT
'Sales Rep',
  `status` ENUM('Active', 'Inactive') DEFAULT 'Active',
  `created_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE `accounts` (
  `id` INT AUTO_INCREMENT PRIMARY KEY,
  `company_name` VARCHAR(150) NOT NULL,
  `industry` VARCHAR(50),
  `phone` VARCHAR(30),
  `website` VARCHAR(100),
  `created_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE `contacts` (
  `id` INT AUTO_INCREMENT PRIMARY KEY,
  `account_id` INT NULL,
  `first_name` VARCHAR(50) NOT NULL,
  `last_name` VARCHAR(50) NOT NULL,
  `email` VARCHAR(100) NOT NULL,
  `phone` VARCHAR(30),
  `assigned_to` INT NULL,
  `created_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (`account_id`) REFERENCES `accounts`(`id`) ON DELETE SET NULL,
  FOREIGN KEY (`assigned_to`) REFERENCES `users`(`id`) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE `leads` (
  `id` INT AUTO_INCREMENT PRIMARY KEY,
  `first_name` VARCHAR(50) NOT NULL,
  `last_name` VARCHAR(50) NOT NULL,
  `email` VARCHAR(100) NOT NULL,
  `phone` VARCHAR(30),
  `company` VARCHAR(100),
  `lead_status` ENUM('New', 'Contacted', 'Qualified', 'Unqualified', 'Lost') DEFAULT
'New',
  `lead_source` VARCHAR(50),
  `lead_value` DECIMAL(12,2) DEFAULT 0.00,
  `assigned_to` INT NULL,
  `created_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (`assigned_to`) REFERENCES `users`(`id`) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

```
CREATE TABLE `deals` (
  `id` INT AUTO_INCREMENT PRIMARY KEY,
  `deal_title` VARCHAR(150) NOT NULL,
  `contact_id` INT NULL,
  `account_id` INT NULL,
  `stage` ENUM('Lead In', 'Meeting Scheduled', 'Proposal Sent', 'Negotiation', 'Closed
Won', 'Closed Lost') DEFAULT 'Lead In',
  `amount` DECIMAL(12,2) NOT NULL DEFAULT 0.00,
  `expected_close_date` DATE,
  `assigned_to` INT NULL,
  `created_at` TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (`contact_id`) REFERENCES `contacts`(`id`) ON DELETE CASCADE,
  FOREIGN KEY (`assigned_to`) REFERENCES `users`(`id`) ON DELETE SET NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

6. Code Implementation & Non-VPS Technical Setup

6.1 Lightweight Native Routing Controller (`index.php`)

To achieve high performance on non-VPS web hosts without heavy frameworks, HA CRM uses a lightweight single-entry controller route:

```
<?php
// HA CRM Core Routing Gateway - Non-VPS Compatible Engine
session_start();
define('HA_CRM_INIT', true);

// 1. Basic Database Connection via PDO
₹host = 'localhost';
₹dbname = 'ha_crm_db';
₹user = 'ha_crm_user';
₹pass = 'SecurePassword123!';

try {
    ₹pdo = new PDO("mysql:host=₹host;dbname=₹dbname;charset=utf8mb4", ₹user, ₹pass, [
        PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
        PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
        PDO::ATTR_EMULATE_PREPARES => false
    ]);
} catch (PDOException ₹e) {
    die(json_encode(['error' => 'Database connection failed: ' . ₹e->getMessage()]));
}

// 2. Authentication Route Guard
₹route = ₹_GET['route'] ?? 'dashboard';
if (!isset(₹_SESSION['user_id']) && ₹route !== 'login') {
    header("Location: index.php?route=login");
    exit();
}

// 3. Lightweight Dynamic Router
switch (₹route) {
    case 'dashboard':
        require_once 'controllers/DashboardController.php';
```

```
            break;
        case 'leads':
            require_once 'controllers/LeadController.php';
            break;
        case 'deals':
            require_once 'controllers/DealController.php';
            break;
        case 'contacts':
            require_once 'controllers/ContactController.php';
            break;
        default:
            http_response_code(404);
            echo "404 - Screen Not Found inside HA CRM";
            break;
}
?>
```

7. Non-Functional Requirements & Security

7.1 Security & Access Rules

- **Data Encryption:** All passwords hashed with `PASSWORD_ARGON2ID` or `PASSWORD_BCRYPT`. SSL/TLS enforced for all client-server traffic.
- **SQL Injection Prevention:** Strict usage of PDO Prepared Statements for 100% of database queries.
- **XSS & CSRF Protection:** All screen output sanitized with `htmlspecialchars()`. Dynamic CSRF tokens injected into all HTML forms and AJAX POST payloads.
- **Role-Based Data Isolation (RBAC):** Sales Reps can only view leads/deals assigned to them or their team. Sales Managers & Admins have full pipeline visibility.

7.2 Non-VPS Performance Tuning

- **Asset Optimization:** CSS/JS assets minified and cached locally. Icons loaded via SVG sprite bundles.
- **Database Indexing:** Indexes applied on `email`, `lead_status`, `stage`, `assigned_to`, and foreign keys to ensure sub-100ms query execution time.
- **Memory Management:** Pagination locked at 25/50/100 records per page to ensure memory consumption remains below 16MB per user request.